Inflation and Financial Policy under the Roman Empire to the Price Edict of 301 A.D.

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INFLATION AND FINANCIAL POLICY UNDER THE ROMAN EMPIRE TO THE PRICE EDICT OF 301 A. D.[1]

1. The inflation and the problem

Some very general quantitative information about the development of the price level in the Roman Empire may be derived from Duncan-Jones who studied prices in the Roman world: "Overall, this evidence suggests that prices in the mid-third century were about three times the level of first-century prices, but that mid-Diocletianic prices were 50–70 times more than those of the first century" (Duncan-Jones 1982, note on page 374/375). Corbier (1985) comes largely to the same conclusion. Duncan-Jones also suggests the use of military pay increases as a crude reflection of inflation (1982, 10). My own study gives some indications about the movements of the general price level in the first century (Wassink, 1989), while analyses of the currency reforms of Aurelian and Diocletian, as we will see, tell something about the extent of inflation in the later third century.

This information combined brings us to the following estimated developments in the price index and the average annual inflation percentages during the first three centuries A. D.

Table: Price level and inflation in the Roman Empire[2]

Date	Reign of Emperor	Estimated price index (denarii, reign Augustus = 100)	Average inflation per annum
27 B.C.– 14 A.D.	Augustus	100	
64 A.D.	Nero	75–85	
C. 200 A.D.	Septimius Severus	200	0,7%
C. 215 A.D.	Caracalla	267	
C. 250 A.D.	Trajan Decius	300	
274 A.D.	Aurelian	700	3,65%
293 A.D.	Diocletian	1400	
301 A.D.	Diocletian	7000	22,9%

[1] Dr. J. P. A. van der Vin (Royal Coin Cabinet, Leiden) was so kind to make comments upon the draft of this study.

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In the graph the development of the price level is illustrated; only the estimation for 215 shows a slight upward deviation from a smooth line. We may conclude that the general price level decreased from Augustus' reign till 64 A. D. and then increased very slowly till c. 250 A. D. Between c. 250 and 293 the growth was higher. An average annual inflation of 3.65% probably does not look alarming, but a period of 43 years is very long for such a moderate inflation. At the end of this interval the yearly inflation will certainly have been above 5%. Huge inflation percentages however occurred in the last years of the century. This hyperinflation must have much embarrassed Emperor Diocletian and his co-regents. It was a new phenomenon for which history did not provide a remedy; it induced them to take emergency measures:

[2] Narrative to the Table:

27 B. C. – 14 A. D.	=	100 –	Base of the indices. Assumed is also that the average index for the first century as a whole is 100.
64 A. D.	=	75 – 85	End of the period of deflation; see paragraph 4 of this study.
c. 200 A. D.	=	200 –	Military pay increased to 1800 or 2000 sestertii. Assume 1800 sestertii per annum, which is twice that of the time of Augustus. The military pay rise during the reign of Domitian is ignored because this Emperor made other economic irrational decisions (see Carradice 1983, 141). Corbier (1985, 105) also states that the prices at the end of the second century were the double of those of the first century.
c. 215 A. D.	=	267 –	Military pay rise to 2400 sestertii per annum.
c. 250 A. D.	=	300 –	Estimation of Duncan-Jones.
274 A. D.	=	700 –	See paragraph 9 of this study.
293 A. D.	=	1400 –	An estimated 500% price increase between 293 and 301; see paragraph 10 of this study.
301 A. D.	=	7000 –	Estimation of Duncan-Jones; 301 having passed the middle of the reign of Diocletian, the maximum estimation is taken.

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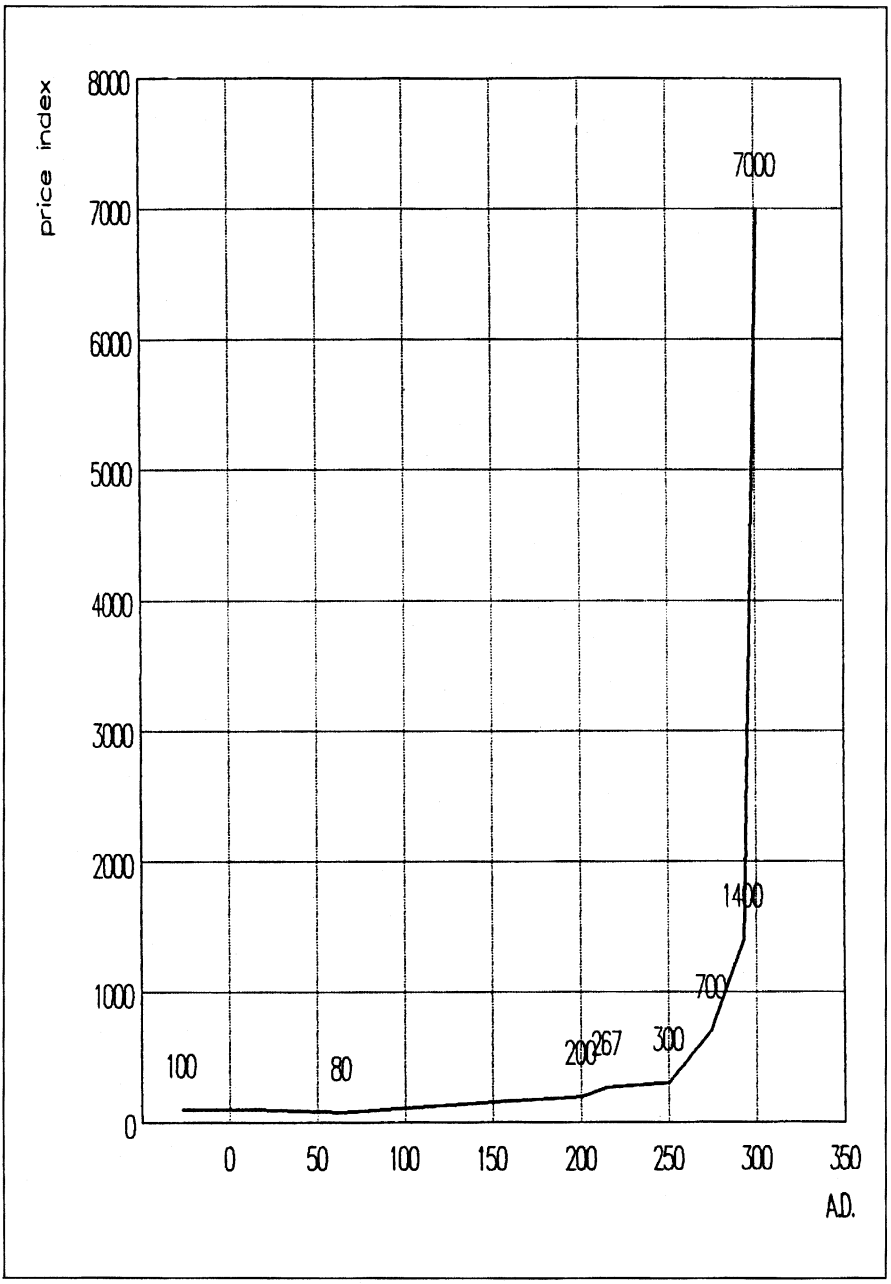

Graph. Development of the price level in the Roman Empire to 301 A.D. (denarii or sestertii; reign Augustus = 100).

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– Probably as soon as 297–298 A. D. they increased the nominal value of the new silver coins, introduced in 293, by 150% in order to prevent that these coins would disappear completely from the circulation because of the fact that their metal value rose above their nominal value. Of this hypothetical retariffing of the silver coins we do not as yet have written or archaeological evidence (paragraph 10 of this study).

As a result of inflation the Emperors were forced again to increase the nominal value of the silver coins as early as the first of September 301. This time they doubled their nominal value in order to keep them in circulation, as we know from an inscription found in 1970 in Aphrodisias.[3]

To prevent a third retariffing because of inflation Diocletian and his co-regents issued in the autumn of 301 the well known Edict of Maximum Prices in which for more than 1200 products and services a legal maximum price is determined. In the Preamble to the Edict they explain its background: prices had risen to an undreamt-of extent; soldiers found themselves forced to pay not just four or eight times the proper price of an article but prices so high as to be fantastic.[4] Such a description confirms the inflation shown on the graph for the years around 300 A. D.: roughly 35% or more per annum!

The cause of the tremendous rise in prices has long since been subject of study. Most authors agree with the Preamble to the Price Edict that it was not a result of a shortage of supplies, but more probably caused by the issue of a flood of – lower standard – coins. This however leaves us with the question why the spending of great quantities of true money by the Emperors Augustus (between 30 and 10 B. C.), Nero (after 64 A. D.) and Septimius Severus (193–211 A. D.) on the contrary had very favourable or at least favourable influences on the economy of the Empire.

The developed price index gives of course a rough approximation of what happened in reality. The actual rate of price increases cannot have been constant and the speed of the price movements is bound to have varied from one region to another. However the price index appears to be a great help in understanding the economic and financial situation and the decisions taken. This study will try to analyse the monetary policy of the Roman Emperors and

[3] Just before 301 A. D. the metal value of an argenteus, a new silver coin issued by Diocletian from 293 onwards of an actual 3.1 grams silver was 56 denarii, if we use the silver price of 6000 denarii to a Roman pound as mentioned in the Edict of Maximum Prices of 301. The nominal value of this argenteus was only 50 denarii. Fragments of another Edict, concerning the nominal value of coins, found in 1970 in Aphrodisias, reveal that the nominal value of the argenteus had been doubled from the 1st of September 301. Other coins were mentioned as well, but from the fragments is not clear which other coins are meant. Ruschenbusch (1977) suggests that these must have been the follis and the old antoninianus, both coins with some silver content.

[4] For the Edict of Maximum Prices see: Graser E. R., 1940, "The Edict of Diocletian on Maximum Prices" in Frank, 1940 (with translation of the Preamble); Lauffer S., 1971, "Diokletians Preisedikt", Berlin and Giacchero M., 1974, "Edictum Diocletiani et Collegarum de pretis rerum venalium I and II", Genoa.

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to retrace some factors that caused the huge inflation of the last years of the third century A. D.

2. The start of the monetary system of Rome[5]

Coinage developed late in Rome. The first coins were probably issued not earlier than 269 B. C. notwithstanding the fact that from prehistoric times onwards the Italian tribes had been accustomed to the use of bronze as a convenient instrument of payment; it circulated in the form of rough lumps (aes rude) of varying weights, shapes and sizes. The earliest Roman coins of around 269 B. C. were bronze Asses and silver didrachms. The As and its divisions (aes grave) were heavy cast pieces of bronze. Coining meant that the rude lumps of bronze of varying weights were replaced by standard pieces bearing a mark of official guarantee. These bronze coins at that time mainly served the internal trade of the native Italian states. The As was then the unit of reckoning. To serve the economic relationship of Rome with its Greek neighbours in southern Italy, a silver coin, on Greek standard, – the didrachm – was struck. Gradually this silver coin was also introduced into the internal Italian trade. This means that Rome became faced with the problem of the relationship between silver and bronze coins. A conversion rate of one didrachm equalling ten Asses is supposed to have existed in this early period. But it is not conceivable that at that time deliberately a bimetallic monetary system was created. Such a double standard would create difficulties. If the conversion rate is officially laid down, the mutual relationship between the prices of the metals involved may at a particular moment change in favour of one metal and then again in favour of the other. In consequence the metal officially overvalued will replace the metal officially undervalued. This phenomenon is known as Gresham's law, which states that "bad money always drives out good money". After the introduction of bronze Asses and silver didrachms the bronze coins were several times reduced in weight, while the didrachm also showed strongly varying metallic values. This is taken to show that the Romans had problems with the two simultaneous coin systems. They will not have completely understood the particular difficulties of a bimetallic standard. We may look upon these early experiences with the coin system as a transition period towards a bimetallic system. For the time being the As remained the most important coin being the unit of reckoning as well. Around 217 B. C. the Romans introduced their own silver coin, the denarius. At the same time the weight of the As was reduced once more and was the conversion rate between silver and bronze coins changed from 1:10 to 1:16. Silver now became the predominant metal. The As, several times reduced in weight

[5] Information derived from Mattingly 1960, Sydenham 1952, Bolin 1958, 131 ff. and Crawford 1985.

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and now tariffed at another rate to the silver coin, lost its position as unit of reckoning to the sestertius, a small silver coin, onefourth of the denarius. Changing the unit of reckoning was apparently not too difficult in those days; probably because coins were not yet in general use. De facto the monetary system became based on silver, but during the Roman Republic the point of making the As a mere token coin never was reached (Mattingly 1960, 27).

Gold coins were not unknown in the Roman Republic. Incidentally they were struck; from 80 B. C. onwards gold coins were only minted at intervals. It was Julius Caesar who, from 47 B. C., when a great scarcity of money existed, tried to mint large quantities of new coins. Forced by a lack of silver he minted gold coins – aurei – on a large scale.

The monetary systems of the Roman Republic were not coherent but only fragmentary; a possible cause of this is that until the first century B. C. money was only in use in a limited sense; many transactions, especially in the daily life of ordinary civilians must have taken place on the basis of barter trade. Nevertheless it is important to note that the Roman coin system started as a bronze based currency with the As as the unit of reckoning but soon developed to a silver based system with a bimetallic tendency and the silver sestertius as the reckoning unit.

A coherent monetary system was introduced in 31 B. C. by Octavian, the later Emperor Augustus, on becoming the leader of the Roman Empire. He introduced a bimetallic coin system, based on gold and silver standard coins with additional token or fiduciary money. These token coins of bronze and copper had an intrinsic value that was below the nominal value. Important for the understanding of later developments was the fact that Augustus replaced the small silver sestertius by a larger bronze coin of the same name. The significance of this modification is due to the following facts:

– A standard silver coin was transformed into a token coin. This fact was probably remembered by Septimius Severus at the end of the second century A. D., when he sharply reduced the silver content of the denarius. He, possibly, intended then to make the denarius a token coin.

– Secondly, the sestertius, now a token coin, remained an important unit of reckoning. Till late in the third century accounts, both public and private, were kept in sestertii; which, though then no longer issued as coins, remained the reckoning unit. Illustrative is a study of Mrozek (1976) who analysed the money amounts mentioned by ten Roman authors of the early Empire. Of 422 amounts mentioned 206 were expressed in sestertii, 111 in denarii, 85 in nummi (nummus strictly means also sestertius) and only 20 (4,7 percent) in aurei.[6]

[6] "Nummus means strictly sestertius, then any coin of gold or silver, finally any coin at all", according to Mattingly in the first issue of his "Roman coins" 1927, note on page 18.

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The Roman monetary system of Augustus comprised the following coins:
- the aureus (appr. 8 grams of pure gold); value 25 denarii
- the denarius (3.9 grams of silver 97.5%); value 4 sestertii
- the sestertius (appr. 27.3 grams of bronze); value 2 dupondii
- the dupondius (appr. 13.65 grams of bronze); value 2 Asses
- the As (appr. 10.92 grams of copper)

In addition there were also the silver half denarius (the quinarius) and the half and the quarter As (the semis and the quadrans). In this way the foundations were laid for the monetary system of the Roman Empire, which functioned almost perfectly over two centuries.

3. The extensive money issues under Augustus[7]

Although in the first century B. C. money was of only limited importance because most transactions took place on the basis of exchanges in kind, a financial crisis occurred in 49 B. C. when Julius Caesar marched to Rome. This crisis manifested itself in a dire need of minted money and resulted in soaring interest rates, whereas prices fell sharply. The shortage of money was a result of the fact that the available quantity of money could not be expanded since a banking system that could supply credit did not exist. The huge demand for coins was the result of the fact that the currency was withdrawn to pay the combating armies, while the public retained such coins as were available in order to feel more secure and keep more possibilities open. Creditors at the same time wanted repayment of their outstanding debts.

Julius Caesar tried to overcome this financial crisis by a number of measures aiming at enlarging the effective quantity of money and also by stimulating the circulation of money. First he minted large quantities of new coins: silver denarii and, by reason of lack of silver, aurei, gold coins then less current (Crawford 1985, 241). No doubt the gold which Caesar carried off from Gaul was used for that purpose. Secondly, a committee was formed to value the real estate of debtors, who could not repay their debts, at prices of before the civil war. Creditors were forced to accept real estate in payment at these prices instead of payment in cash. This meant in fact that real estate took the form of substitute money, with the assumption that prices would soon return to pre-war levels. Thirdly, Caesar decreed that nobody was allowed to keep more than 60,000 sestertii in cash; a measure probably based on older legislation.

As a result of these and other measures and of the gradual disappearance of the violence of war from Italy itself, the financial crisis eventually faded away. However, the demand for coins persisted, the more so since during the crisis people had learned to appreciate the convenience of possessing it. The civil

[7] See: Frederiksen 1966; Frank 1935 and 1940; Nicolet 1971; Pekary 1976 and Thornton 1975.

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war had also enlarged contacts with eastern countries, where the transition to a money economy had advanced further. These contacts also will have stimulated a desire to use coins for more economic transactions.

In the beginning of Augustus' reign Rome was still troubled by this constant demand for more money, which had taken the form of a deflationary depression. Faced with this the Emperor proceeded by taking the appropriate step to counteract it: the spending of huge sums of money. This was done directly by means of donations (congiaria), by executing many public works and by providing free public entertainment. It was done indirectly by the distribution of purchased grain and the donation of purchased land to former soldiers.

This spending of large sums of money lasted for about twenty years, i. e. till about 10 B. C.[8] This was obviously possible thanks to the treasures that Augustus had carried off from Egypt, while the production of gold and silver mines also had recommenced. After 10 B. C. considerably less money was issued. Apparently the extra means the Emperor had at his disposal became exhausted.

The question remains whether Augustus started his policy of spending huge sums of money only to counterbalance the deflation, in which case he may have had already a notion of the principles presented by John Maynard Keynes during the great depression of the thirties of this century (Keynes 1935). It is however also possible that Augustus started a redistribution of wealth acquired not only by the exertions of Emperor himself, but also by all soldiers and civil servants of the Empire. Augustus soon experienced the favourable influence of his policy; as a direct result of the abundant issues of money the interest rate fell from 12% to 4% and the prices of landed property rose. Deflation was neutralized; gradually sufficient money came into circulation for the barter economy to transform itself into a money economy. Since in a money economy trading is much easier than in a barter economy, numerous new economic possibilities were created and many unemployed civilians found jobs. Their prosperity increased. They in turn spent their wages, which provided farmers and craftsmen with new possibilities. When eventually the liquidity was sufficient it was possible to determine the taxes payable by civilians in money. This obliged taxpayers to market their products or to offer them for sale to the army to be able to pay their taxes. From this a new economic stimulus proceeded, which would not have been there if taxes had been collected in kind (Hopkins 1980). In short the multiplier effect proceeding from the extensive issues of money resulted in huge economic progress.

[8] The year 10 B. C. is among others mentioned by Frank 1935, 337. Thornton 1986 studied the building activities around Rome and concluded there had been a striking peak between 11 B. C. and 4 B. C. "In about 2 B. C., however, something happened: the Emperor suddenly stopped building". The conclusion may be drawn that the year 10 B. C. as final year of Augustus' spending period is perhaps a little to early.

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It did not result in inflation because sufficient factors of production, in the first place labour und land, were now, after the long civil wars, available to the economic process.

Probably Augustus was not acquainted with economic laws, but his financial and economic policies were so accurate and successful that they must have been an ideal example to the Emperors of the Roman Empire of the following centuries.

In the latter half of Emperor Augustus' reign, i. e. after 10 B. C., the issues of money decreased considerably. By then the Empire was provided with a fairly large effective quantity of money. We lack the quantitative information which allows us to be very exact on this point, but it is clear that in the course of a few years a deflationary process had set in again. This was due to the fact that through the persistent economic growth and urbanization the existing quantity of money in the Empire became gradually too small. The deflation which occurred after 10 B. C. as a result of a relative shortage of money manifested itself by falling prices and rising interest rates. This deflation caused considerable tension in the Roman Empire of the first century, since falling prices meant a substantial increase in the burden of taxation on civilians. As far as the state was concerned falling prices led to higher spending since soldiers' pay, the main item on the state's budget, could not be lowered. Obviously the rise in interest rates burdened debtors especially, and in the Roman Empire at first these were mainly the young ambitious politicians, who had borrowed money for their election campaigns. The tension of the deflation resulted, among other events, in an insurrection which broke out in Gaul, during the reign of Tiberius in 21 A. D. (Christopherson 1968), in an interesting financial crisis in 33 A. D., also under Tiberius (Frank 1935) and in public unrest under Nero in 61 and 62.

4. The great money issues under Nero[9]

The public unrest of 61 and 62, which may be due to the situation of deflationary depression, induced Nero to assume control of the Empire's financial and economic policy himself. Before 62 this was in the hands of the governors Seneca and Burrus.

Nero must have been aware of the fact that Augustus, ninety years earlier, had successfully counteracted a situation of deflation by improving the currency system and by spending huge amounts of money. Therefore from 62 Nero started to copy Augustus' financial policy in all its aspects. In 64 he reformed the monetary system; it looks as if the relations between values and prices, also those between silver, gold and copper were restored to the early

[9] See: Thornton 1975 and Lo Cascio 1981.

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Augustan level. In order to correct the reduced general level of prices Nero lowered the intrinsic value of the coin standard: the denarius by over 15% and the aureus by 10%. In doing so he restored simultaneously the altered relation between the values of gold and silver.

From the devaluation of the denarius by 15% may be derived that the general price level at 64 was at least 15% lower than in the reign of Augustus. At least 15%, because it is possible that Nero limited the devaluation to 15% to prevent that in accordance with Gresham's law all older denarii would disappear immediately out of circulation. In other words it may be that 15% devaluation for silver coins was thought to be the maximum possible.[10] The actual general price level in 64 may therefore have been more than 15%, say 25% lower than in the reign of Augustus. The devaluation was effected by a reduction in the weight of the denarius from 3.9 to 3.4 grams and a reduction of the silver content from 97.5% to 93.5%.[11] The aureus, which had an average weight of 8 grams, was reduced in weight to about 7.2 grams. Oddly enough simultaneously the weight of the minor coins was raised a little (Sutherland 1976, 126). No doubt the price of copper had fallen in relation to silver and gold since the days of Augustus, which is not surprising in view of the reduced general price level.

From 62 onwards Nero spent huge amounts of money for periodic distributions of grain and money among the public, the provision of impressive public spectacles and above all for an extensive construction and building programme. In this programme special attention was paid to the digging of canals and it is not inconceivable that the notorious great fire of Rome in 64 fulfilled a purpose in the construction policy designed by Nero. These substantial money issues were possible thanks to a well filled treasure and the boosted production of the Spanish silvermines, which had already been nationalized under Tiberius. According to Pliny the gold production of the Dalmatic mines was also considerable in Nero's days, while Tacitus informs us that in those days also gold from the temples was used to finance the state's expenditures.[12]

Nero's new policy was successful; prosperity suddenly increased enormously. The reform of the monetary system supported the large public expenditures, which functioned as a kind of deficit spending, i. e. the higher demand for goods increased prosperity and did not lead to inflation because production factors, especially labour and land were still sufficiently available due to the past period of deflation which lasted more than seventy years.

[10] It may also be possible that the limitation of the scope of the devaluation was dictated by that of the aureus: 10%. Also Caracalla devaluated the aureus by only 10% when he wanted to restore the disturbed equilibrium between the value of the aureus and that of the denarius (paragraph 7).

[11] All percentages of silver fineness have been borrowed from Walker 1976, 1977 and 1978.

[12] Tacitus, Annales XV 45.

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The risks of a deflationary policy were clearly understood by then, and the necessity to pursue a policy aiming at the creation of employment was generally accepted. Shortages in liquidity can no longer be traced in the decades after Nero. Besides, based on the number of types of coins known, we have the impression that Nero's successors, too, had coins minted and circulated on a large scale.

5. The problem of maintaining bimetallism

The problem rulers were especially confronted with in the period of more than one hundred years after Nero was that of the bimetallic nature of the coin system: the problem formulated by Gresham. In the two first centuries A. D. the problem of the bimetallic Roman monetary system was caused by a rise in the price of silver compared to other goods. This price increase was both a consequence of the great demand for silver on behalf of the Roman monetary system, and of the disappearance of much silver across the borders, mainly to free Germania where silver was preeminently the commodity in which wealth was secured and hoarded. In this period silver also rose in value with regard to gold, therefore Nero had devalued the denarius more than the aureus. The relation between the prices of gold and silver at the time of the Roman Empire can be worked out at intervals. It developed approximately thus:

Period	**Value of bullion silver in proportion to the value of bullion gold.**[13]
Augustus (27 B. C.–14 A. D.)	8 %
Nero (54–69)	8.5%
Marcus Aurelius (161–180)	11.4%
Diocletian's Price Edict (301)	8.3%
Beginning 4th century	6.7%

It seems that the price of silver was low in proportion to the price of gold in the early days of the Empire and that it rose afterwards, especially when Trajan had conquered Dacia, which made a large war booty of gold bullion available. At the end of the second century the price of silver dropped again in proportion to the price of gold. From Nero's day onwards the influence of the rising silver price was compensated for by reducing the silver content of the denarius (Bolin 1958, 208ff.). The development of the silver content up to the time of Marcus Aurelius was thus:

[13] See also West 1941, note on page 3 and Bolin 1958, 265, 313.

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Period	Average silver content
Augustus (27 B. C.–14 A. D.)	97.5%
Nero (from 64)	93.5%
Trajan (from 107)	87 – 90 %
Antonius Pius (from 148)	82 – 85 %
Marcus Aurelius (from 179)	74 – 77 %

It should be mentioned here that after Nero fineness reductions did not take place officially, since during the reign of almost all Emperors of the first and second century also denarii with a content of 92–93% were minted. But more and more coins with too low a fineness were minted, and on top of this the minimum content became increasingly lower. Through this somewhat obscure, but very effective policy the bimetallic system was kept sound. Some authors point out moreover that the Roman Emperors also controlled the gold and silver mines and so the supply of gold and silver. In this way they were able to regulate the prices of the metals to a certain extent (Lo Cascio 1981, 78). It is remarkable that beside the necessary reduction of the fineness of the denarius, the aureus remained unaltered. This means that the fineness of the silver coin, the reckoning unit, was manipulated in order to maintain the position of the aureus. As a consequence of this policy the aureus became de facto the major coin of the imperial monetary system, in spite of the fact that the denarius maintained its position as a unit of reckoning. This divergency would ultimately become a major cause of the inflation of the third century A. D. The gradual decrease of the silver content, which started under Nero, did not lead to the rapid disappearance from circulation of coins with a higher fineness. This was partly a result of the fact that the decrease in content imposed by the authorities at first took place gradually and obscurely. Besides, the minimal difference in value between minted silver and silver bullion acted as a buffer. Also the older coins with an averagely higher fineness had already become a little worn, which somewhat reduced the differences in value with the new coins.

The reductions in silver fineness should not be looked upon as a trick to make more money, as is often thought, but as a purposely selected policy to maintain the monetary system. This also appears from the fact that the years in which content reductions took place, 107, 148, und to a lesser extent 179, were not characterized by remarkable financial problems, which might have induced the Emperor to tamper with the coin. Neither was the standard of the gold coin ever meddled with after Nero till 193, though it would have been an obvious action, had an Emperor's own short-term interest been the object.

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6. The monetary policy of Septimius Severus

After a long period of economic stability the Roman world was confronted under the reign of Marcus Aurelius (161–180) with a number of interrelated problems: invasions of the barbarians, the outbreak of the plague of which a considerable part of the population died, a declining economy and deficits of the Roman treasurer.

The declining economy, probably in concurrence with the sharply stepped-up production of the Riotinto silver mines, caused a fall in the price of silver, also in proportion to gold (Jones 1980). This was a new phenomenon, since from the days of Augustus the price of silver had risen. The Emperors had learnt meanwhile how the bimetallic system should be protected against a relative rise in the price of silver, but what were they to do in case of a fall in the silver price? It seems self-evident that in this case the silver content should be raised. But it should be borne in mind that Gresham's law would cause the clearly recognizable, brand-new high-grade coins to be withdrawn from circulation immediately and hoarded or melted down.

It is not known when the price of silver started to fall, but the fact that the silver content of the denarius was raised a little, though not clearly perceivable, under Marcus Aurelius between 166 and 170 may have been a clue that then for the first time, and probably only temporarily, there was a significant fall in the price of silver (Walker 1977, 60). However, between 170 and 192, the average silver content of the denarius was gradually reduced again. It must be doubted however, whether this reduction, especially during Commodus' reign, still had a rational basis. After all, at the end of Commodus' reign there must have been a wide discrepancy between the intrinsic value of the denarius and the $1/25^{th}$ part of the aureus. This is very clear from the fact that almost all successive Emperors and pretenders to the throne took economic measures to restore the monetary system to a sound basis as soon as they had an opportunity to do so, especially with regard to the ratio one aureus equals 25 denarii.

Pertinax, Didius Julianus and Clodius Albinus all attempted to solve the problem of the ratio gone awry by raising the silver content of the denarius or by reducing the standard of the aureus (Walker 1978, 61; Bolin 1958, 191). The latter might have been an effective measure if it had concerned only a slight disruption, but with a wider discrepancy Gresham's law would soon be activated and older coins of larger value would rapidly disappear from the circulation.

Why the Emperors thought it so important to restore this ratio and why they were not prepared to accept another conversion rate is not clear. But supposedly the ratio one aureus equals 25 denarii was so fixed in laws, regulations, contracts, and customs that it was considered impossible to deviate from it. However, during Commodus' reign a different conversion rate must already

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have been used unofficially, which must have been regarded as undesirable. Septimius Severus (193–211), the eventual victor in the struggle for the imperial throne, tried to combat the financial and economic problems with the success formula of Augustus and Nero: reorganising the currency system in order to make it healthy and coherent and issuing extensive quantities of money in order to stimulate the economy.

As a first step he reduced the silver content of the denarius from 72–73% as applied between 190–192 by Commodus to 50–60% fine. This means that he brought the fineness to the lowest possible level. Based upon the assumption that the relative value of silver and copper during the whole Roman period remained constant at about 100:1 (Bolin 1958, 303), one may calculate that with a fineness of less than 50.9% theoretically the intrinsic value of the denarius becomes less than 16 times the copper value of the important minor coin, the As. This would inevitably result in a disappearance of the As from the money circulation. This explains why the fineness of the denarius under Septimius Severus' successor Caracalla (211–217) also remained 51% or slightly higher. A rational explanation for Septimius Severus' policy to lower the fineness of the denarius as far as possible is that he may have wanted to downgrade the denarius from a standard coin to a fiduciary token coin, just as Augustus had done with the silver sestertius, by replacing it by the bronze coin of the same name. Such a policy would be very understandable. Already during one century and a half the Empire had been faced with the problem of the bimetallic system, mainly as a result of the permanent increases in the price of silver. This development was counterbalanced by a gradual decrease of the fineness of the denarius, while the gold content of the aureus was left unchanged. A move to the golden standard, by making the silver denarius a token coin, seems under the existing conditions to have been very rational; the more so since the relation between the values of aureus and denarius had irretrievably gone awry. There is an indication that Septimius wanted to enforce the conversion rate of the new token coin, the denarius at one aureus equals 25 denarii equals 400 Asses by law. An inscription has been found in Mylasa in Asia Minor dating from 207–210, in which the confiscation was announced of aurei exchanged otherwise than with the authorities for that purpose. If at the unofficial exchange more denarii than 25 were given for one aureus, additional penalties were imposed, which benefited the imperial treasury as well as the informer and the judge. This concerns an imperial order which, probably, had to be executed by local authorities (Pekary 1959, 464). Secondly, Septimius Severus issued extensive quantities of coins. This can roughly be illustrated by the number of types of coin that were struck. In the first three years of his reign Septimius issued 114 different types of denarii per annum, while for example Antoninus Pius issued on average only 17 types per year (Pekary 1959, 457). After 198 aurei were minted on a large scale; at any

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rate a considerable increase in the number of aurei types can be observed. This no doubt was a result of the fact that in 198, during the war against the Parths the large royal treasure, mainly consisting of gold, was seized.

The large issues of money were spent for a large part on the army; the soldiers received for the first time in over a century a considerable rise in pay. Also Septimius added three new legions to his army. In addition to the customary distributions of grain, he indulged he populace of Rome with six congiaria of an estimated total of 220 million denarii, as well as with extravagant games and free medicine for the poor (Cary and Scullard 1975, 499). Well known is also the building programme of the Emperor. At Rome he built a new imperial palace on the Palatine, he adorned the west end of the Forum with his arch and he began the construction of a vast suite of baths. In the provinces he executed a huge construction programme in his African hometown Leptis Magna, and spent large amounts on road repairs and took upon himself the costs of the postal service, which had hitherto rested on the shoulders of the wayside municipalities.

The large issues of money at first certainly had a favourable influence on the economy. The interest rate decreased from 6% to 4%. The prices did not rise significantly (Corbier 1978, 277/278), but no doubt the prices were tempered by the increase in the gold stock-seized from the Parths – and in the quantity of gold coin which stopped the relative fall in the price of silver and in the value of the denarius. These developments clearly show that the value of the coins only to a certain extent was determined by the value of their content of precious metal (Bolin 1958, 99).

In general Septimius Severus' money spending policy is deemed favourably. It led to automatic increases in taxation in the period of renewed material prosperity and restored the economic setback due to the great plague of Marcus Aurelius' days and the misgovernment of Commodus. The new money issues mobilized the newly available production factors, labour and soil (Cary and Scullard 1975, 499).

Notwithstanding his economic success Septimius Severus did not succeed in turning the denarius into a token coin. Apparently, confidence was lacking, which is not surprising, as the value of the denarius had long since fallen to less than $1/25^{th}$ of the aureus. Perhaps also historically the denarius was too important a coin to be degraded to a minor coin and it was still, together with the sestertius the unit of reckoning of the nation. However de facto an end was made to the bimetallic currency system, but not with the denarius as a token coin as Septimius intended. The value of the denarius was no longer coupled to the aureus.

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7. The failure of Caracalla's monetary reform

After Septimius Severus' death in 211 there was still a strong tension between the intrinsic value of the denarius and that of the aureus. In practice this meant, as we have seen, that much more than 25 denarii had to be paid for one aureus. At the same time the fineness of the denarius was at its lowest possible point: 51%. A further reduction would push the bronze As out of circulation.

Septimius had tried to solve the currency problem by transforming the denarius from a standard coin into a token coin. But he had failed; even his coercion measures were insufficient to have the denarius accepted as a fiduciary coin.

Although no written records have come down to us about this operation, it seems if Septimius Severus' son and successor Caracalla made in 215 another great effort to implement a logical reorganization, which was now to restore the old equilibrium one aureus equals 25 denarii. This reorganization comprised:

- The reduction of the weight of the aureus from 7.2 to 6.5 grams. This weight reduction was apparently the largest that could be realized without Gresham's law causing all the old aurei to be withdrawn from circulation.
- The introduction of a "new denarius", which was clearly distinguishable from the old one, with an intrinsic value of $1/25^{th}$ of the aureus reduced in weight. This new coin had a weight of a little over 5 grams with a silver fineness maintained at 51%. The intention must have been to withdraw the old denarius from circulation and to exchange it for the new one.

As Mickwitz assumed as early as 1932 the new denarius, nowadays called antoninianus, will have been introduced as $1/25^{th}$ aureus (Mickwitz 1932, 34). We do not know the Roman name of the coin, but in all probability it did not have a name of its own because it was to be called denarius! Here a comparison with the situation in France, where after the 1958 coin reform people talked of old and new francs for some time, urges itself upon us.

It is surprising to see that as early as 215 a coin reform was conceived that strikes us as modern. Unfortunately the attempt was not crowned with success and the consequences of the failure were far-reaching indeed. A further reduction of the intrinsic value of the denarius causing an enlarged discrepancy between the value of the denarius and the aureus and problems for the As, which now rapidly disappeared from the circulation, must have seriously injured the monetary system. However, at the same time the monetary volume continued to grow under Caracalla (Corbier 1978, 277).

The question must be posed why Caracalla's attempt to reform the monetary system ended in failure. In this respect it is important to know that a coin reform is only possible if the old coins in circulation can be replaced by new ones at short notice; if not, both coins will start circulating simultaneously at a

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certain conversion rate. This must have been the case here and in this situation the new denarius will have obtained twice the value of the old one. A logical conversion rate if we remember that in the Roman coin system:

2 quadrantes	equal	1 semis
2 semisses	equal	1 As
2 Asses	equal	1 dupondius
2 dupondii	equal	1 sestertius
2 sestertii	equal	1 quinarius
2 quinarii	equal	1 denarius

Caracalla's reform could only have been a success if the operation had been completed within a couple of months and been accompanied by sufficient information to the general public. Unfortunately Caracalla lacked the technical possibilities and the time – he was murdered in 217 – to finalize his reorganization successfully. That the introduction of the new denarius, the antoninianus, was looked upon as a failure by contemporaries is clear from the fact that Caracalla's successor Macrinus, who only ruled fourteen months, immediately stopped production in 217. The old denarius was minted again, and this time with a somewhat higher silver content, viz. 50–60%, than Caracalla had used (51%). So Macrinus obviously wanted to cancel out Caracalla's mistakes by returning to the situation under Septimius Severus. Macrinus did maintain the reduced weight of the aureus.

After 218 all hope that the ratio one aureus equals 25 denarii could ever be restored must have been abandoned. Both coins must have circulated simultaneously without a fixed conversion rate: a situation that unofficially must have existed since c. 190.

Under Macrinus' successor Elagabalus (218–222) the antoninianus was minted again occasionally in the years 218–219. This coin, circulating as a double denarius, must at that time still have been relatively scarce. But being inferior to the denarius it must have had a negative influence on the circulation of the denarius, according to Gresham's law. This must have been the reason that coinage of the antoninianus again ceased in 219.

But damage was already done to the coin system: the circulation of denarii was limited, because people were inclined to hoard these coins as long as there were antoniniani available. Also the As vanished from the money circulation, but this minor coin had lost part of the importance it had under the early Empire due to the fact that the general price level was now more than 2.5 times that of the time of Emperor Augustus. On the other hand the favourable influence of the increase of the money volume under Septimius Severus and to a lesser extent under Caracalla still endured for some time.

It was the financial crisis of 238 that brought a major change in the monetary situation of the Empire.

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8. The monetary and financial crisis of 238 A. D.

It is surprising that the prices in the Empire did not rise significantly after the debasement of the denarius under Septimius Severus, the extensive money issues under Septimius and Caracalla and the failure of the coin reform of 215 that left the country with an inferior double denarius, the antoninianus. The annual inflation remained extremely low to our view: still under 1%. As explained, this was partly due to the considerable economic growth resulting from Septimius' money issues that mobilized available production factors. On the other hand one must keep in mind that society was still very traditional; price rises just did not exist. The economic situation remained stable for the time being. This changed in 238 and the evidence suggests that the changes of that year spelled the beginning of the end of the imperial currency system. In that year the money in the treasury was insufficient to meet immediate expenses such as the payment of the army. The army of course was the basis of the government's power. The total absence of any concept of state credit forced the Emperor to debasement of the currency. This debasement of 238 was unique because for the first time in Roman history it was done without a defensible monetary policy. The debasement comprised:

- from 238 on antoniniani were minted almost exclusively;
- at the same time the weights of the antoniniani and aurei were reduced;
- the fineness of the silver coins was reduced to c. 48% (Walker 1978, 68, 138).

The most serious consequence of this monetary reform was that through the effects of Gresham's law, within a few years the denarius vanished entirely from the circulation. Immediately after the 238 re-introduction of the antoniniani the government must have experienced the fact that the denarius was forced out of circulation. Late in 240, extending to early 241, a brief attempt to halt this unwanted development was made when the denarius as a denomination was reintroduced (Walker 1978, 68), but undoubtedly it was too late to save the denarius.

The Empire was deprived of its main coin, a unit of reckoning for at least three hundred years! The disappearance of the denarius must have had a tremendous influence on the effective quantity of money in the Empire. By a rough estimate the reduction of the money volume within a short time amounted to at least 50%. The need to mint new antoniniani became pressing, which was only possible by reducing the fineness of the coin further; Walker (1978, 58) records for 253 a percentage of 35. Analyses of coins minted in 270 show an average fineness of not quite 2.5% silver (Bolin 1958, 249).

The continual reductions of the silver content of the antoninianus also caused the remaining minor coins, the sestertius and the dupondius, to be withdrawn from circulation before 270. In this way the bronze token money as well as the denarius disappeared from circulation by the beginning of the

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second half of the third century, and the monetary system of the early Roman Empire collapsed. After over two and a half centuries this society was thrown back from a developed monetary economy to one that for a large part had to make do with barter trade, since only the antoninianus, which was devaluating quickly, and a limited number of aurei still circulated independently as instruments of payment. So within a few decades the effective quantity of money had been decimated.

It is hardly conceivable how sharp the decline in prosperity must have been in those years. Trade and transport, and in their wake many other forms of economic activity must have been wiped out. Many people must have lost their living within only a few years.

The sharp reduction in the effective quantity of money that occurred immediately after 238 must have generated a strong deflationary effect, which means decreasing prices and increasing interest rates. On the other hand, the diminishing fineness of the antoninianus after 238 decreased the actual value of this new main coin of the Empire. This decrease in value meant an increase in prices. The existence after 238 of these two influences on the general price level explains that the price level in the Empire did not increase much till the middle of the century. At that time the deflationary tendencies lost their force while the fineness reductions of the silver coin went on. Finally, the prices started to rise; at first by more than 1% per annum, then more, finally ending up in what we call nowadays a hyperinflation.

9. Requisitions in kind and the financial reform of Aurelian

Around the middle of the third century Gresham's law forced the denarius, and what was left of the bronze minor coins, out of circulation. Only the antoninianus remained in use because the aureus was at that time not really a coin anymore. Fluctuations in the weights of aurei, at least after Trajan Decius (249–251), give the impression that these coins ceased to have a uniform metallic value and a fixed place in the system of coinage (Bolin 1958, 263).

The sharp diminution of the effective quantity of money decimated the monetary economy and forced an increase in barter trade, with negative consequences. One of these was that it became very difficult for the state to collect the old money taxes and brought about a move to the collection of taxes and the payment of the army largely in kind. The "indictiones extraordinariae" in practice became the most important form of revenue after the middle of the third century, but at first the burden was no doubt very incidentally placed and also unequally distributed between provinces, cities and individuals (Jones 1964, 46, 61). Not only the requisitions were unequally distributed but also they shut a large part of the population out from contacts with the market and the market economy: these people now had to work

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almost exclusively for their own food and clothing and for the governmental requisitions. They did not receive money for their products and could not spend it for the products of the urban craftsmen, who in turn suffered. Craftsmen and in consequence traders disappeared gradually from the cities, which obliged Emperors to set up state armourers and cloth factories to furnish their armies. They had perforce to employ compulsion to keep up the supply service of Rome. As a consequence the opportunities for the private trader sharply declined (Cary and Scullard 1975, 537). Voluntary service, which flourished under the Principate gave way to a system of compulsion which limited the individual's right of choice. It meant that there came an end to the advantages of the division of labour which brought about the prosperity of the early Empire. This ultimately resulted in poverty and social injustice.

Of course the government noted these unfavourable developments and must have realized that these were the consequences of a decimated effective volume of money. Coining new money was necessary but the old silver denarii had vanished in the hoards of the general public and new silver was scarcely available. The only possibility left was to drastically reduce the fineness of the antoninianus yet again. Around 270 huge quantities of antoniniani were struck with a silver content of less than 2.5%. The workers in the state factories were probably paid with these debased antoniniani which was no doubt considered as the reason of their poverty and that of the whole population.

This situation resulted in 271 in rebellion of the workers of the mint at Rome. People demanded that confidence in the coins issued by the state should be reestablished (Kienast 1974, 565). The uprising may have induced the Emperor Aurelian to reform the currency system in 274. He issued a series of new coins. Most important was a new silver coin of 3.9 grams, similar to the antoninianus issued by Gordianus III in 238 at the final introduction of that coin.[14] The reverse of the new coin bore the marks either XXI in the latin speaking part of the Empire or KA in the greek speaking part. The coin, though subject to several minor abortive reforms survived until the reform of Diocletian of 293, which introduced the follis to replace Aurelian's coin and the older antoninianus as the basic coin of the imperial monetary system (Harl 1985, 268). MacMullen (1976, 110) calls the new Aurelian coin "aurelianus", because it is no longer an antoninianus; a suggestion which merits copying. That Gresham's law left the aurelianus in circulation means that it must have had a much higher nominal value than that of the antoninianus (2 denarii). A quick analysis of the price developments as given in paragraph 1 shows that the prices in 270 were about two and half times those of 238. This makes it acceptable that the nominal value of the aurelianus, resembling the early

[14] The weight of 3.9 grams goes back to the weight of the denarius of Augustus.

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antoninianus of Gordianus III, was five denarii.[15] It confirms the suggestion that the mark XXI means 20 sestertii (5 denarii) for 1 new silver coin as made by Sutherland (1961). Based on this confirmation we accept for 274 a price level index of 700 that is 2.5 times that of 238. An interesting fact about the aurelianus is the declaration of value on it. Government had come to grips, however unsuccessfully, with the idea of a universal fiduciary system (MacMullen 1976, 110).

Aurelian also introduced, in 274, a new gold coin of 4.6 grams, a new sestertius, and a double sestertius.

By introducing the new coins the Emperor tried to restore public credit. He delivered new money in exchange for bad, which the people were commanded to bring to the treasury. This monetary reform was again a failure. Just as his predecessors Aurelian had too little precious metals to give the exchange of coins the necessary volume. The exchange was probably only limited to Rome (Kienast 1974, 547 note). Also Aurelian's time was too short, as he was murdered in 275. The Empire was left in a poor condition, the monetary system was far from sufficient and soon the debasement of the aurelianus began. But the army had to be fed and clothed. The government requisitioned without payment the wheat, meat, wine, oil, textiles and leather needed to feed and clothe the troops and issued them free. The effect of this development, eliminating the currency from a large and important range of transactions, was an increase in inflation: the slowly rising mass of inferior currency was placed against a smaller quantity of goods.

The whole burden of taxation, apart from customs dues, which were insignificant, fell on the rural population. An organized requisition in kind soon, and certainly under Diocletian, meant that it became illegal for peasants to leave their registered domicile. In effect the tenant was bound to his particular farm and thus to his landlord (Jones 1964, 68, 69). Before then Aurelian and other rulers had already issued ordinances calling upon the cities of the Empire to find cultivators for then neglected fields. This shows that the problem of the derelict lands could no longer be remedied by the time-honoured device of colonization (Cary and Scullard 1975, 537). Agricultural population became hereditary tied to the soil. This hereditary caste system extended gradually; Diocletian compelled the sons of soldiers to serve in the army, while the workers in the mints and the weaving and dyeing factories which Diocletian created were state slaves and their service also hereditary. Goldminers and washers too were later a hereditary caste. Other hereditary classes which probably already existed in Diocletian's days were the guilds of bakers,

[15] We lack information on the fineness of the aurelianus to be more exact. The coin had a silver wash and probably 4% silver. The information used of Kienast 1974 and other authors is that the coin was similar to the antoninianus of Gordianus III. This means that there was a 150% inflation between 238 and 274 A. D.

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butchers, bargees, carters and so forth in Rome and the diocesan guilds of skippers (navicularii), who carried corn and other public cargoes to Rome (Jones 1966, 35, 36). Requisitions in kind meant that some producers did not participate in a monetary economy. Making these producers to a hereditary caste meant that they could not participate any more, even if they wanted, in the free economic process. It became very difficult for them to react to economic stimulus like money injections through public spending programmes. The part of the economy which was monetized was still further reduced.

10. Diocletian's economic policy

When Diocletian became Emperor in 284 the Empire had already suffered anarchy and chaos for half a century. The economic situation was very poor. The people were impoverished and partly bound in hereditary castes. Taxes were levied in kind in a manner that caused much injustice. The only circulating coins were the old antoniniani and the aureliani, most of them very much debased, and the aurei, which were, as we have seen, no longer real coins due to the fluctuations in their weights. Inflation was higher than ever; at least 5% in 284 according to the graph in paragraph 1 of this study.

The first great service Diocletian did the Empire was rationalization of the requisitions in kind by making them an annual event and by distributing the burden equally between provinces, cities and individuals. To provide a basis for the new assessments a series of censuses was held throughout the Empire. The calculated rate of levy was published annually. It is the first example in history of a yearly budget whereby the taxes were adjusted to the estimated needs of the government (Jones 1964, 66). There is evidence that Diocletian tried to follow the great examples of Emperors like Augustus, Nero and Septimius Severus by planning a thorough reform of the monetary system, to be followed by extensive spending. Already from the beginning of his reign Diocletian resumed, after a gap of many years, the regular issue of gold coins. They were marked with their weight, which was at first 70 to the Roman pound (4.6 grams), but later 60 (5.35 grams). This change in weight can only be explained through the fact that the Emperor wanted to introduce a coherent new currency system with a new fixed exchange rate between gold and silver coins. In other words he intended to reintroduce a bimetallic currency system. Probably the new gold coin was called solidus, at least this name is written on the fragments of the Price Edict that were found in 1970 in Aezani (Crawford, Reynolds 1975). It emphasizes that Diocletian wanted to make a complete new start with his currency system. A new denomination (follis) with a new design, silver washed copper, 10.52 grams, was introduced. It was struck in great quantities and was essentially the "peoples coin", mass produced in all the fourteen mints of the Empire for worldwide use. Now some types of this

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coin were marked XXI (Bolin 1958, 296), which means that the follis received the nominal value of five denarii. Its main importance lay in the provision once more of a fairly abundant coinage struck to an exact standard. The existing aurelianus, debased since its introduction in 274 to a copper nummus was retariffed, probably to two denarii (Bolin 1958, 195).

Diocletian continued to strike a coin of the same standard and obverse design as the aurelianus, but now no longer bearing the mark XXI. This nummus, together with what was left over of the older aureliani and antoniniani through the working of Gresham's law, probably formed a coin of two denarii.

Diocletian resumed the minting of a silver coin with the same metallic value as the denarius of Nero. Another example of how much Roman Emperors were inclined to use history in developing their policy! Generally is assumed that the new coin (argenteus) in 293 had a nominal value of 25 denarii (Harl 1985, 263).[16] However it seems more logical that it was 20 denarii to the argenteus, because the coin was retariffed in 301, as we will see, according to an inscription at Aphrodisias, from 50 to 100 denarii, while another coin, which must be the follis, was retariffed from 12.5 to 25 denarii. Between 293 and 301 another hitherto unknown change of 150% in the nominal values of both coins must have taken place. If the argenteus was not worth 20 denarii in 293 the retariffing sometime between 293 and 301 of this coin and of the follis, must have been unequal, which is hard to imagine. In the Edict of Maximum Prices of 301 the price relation of gold versus silver is 72 000 : 6000 = 12 : 1.

Assuming this relation already existed in 293, it is easy to calculate that Diocletian intended a solidus to be 20 argentei.[17]

All this means that Diocletian tried in 293 to introduce the following coherent bimetallic currency system:

1 solidus equals		20 argentei
1 argenteus equals	4 folles	= 20 denarii
1 follis equals	5 denarii	= 20 sestertii
1 new copper nummus (and older antoniniani and aureliani)		= 2 denarii

[16] The assumption that an argenteus in 293 equalled 25 denarii probably goes back to the analysis of Mickwitz 1932, 68 of which Bolin 1958, 304 writes: "Mickwitz' conclusion, which he draws with considerable hesitation is that there must have existed a 25 denarii piece, and he identifies it with the silver coin introduced by Diocletian." A conclusion based on the fact that the prices in the Edict of Prices are often divisible by 25. The fact that the Augustian aureus was worth 25 denarii may also have played a role. As we have seen later Emperors often copied situations and facts of the early Empire (Jahn 1975, 96).

[17] One solidus of Diocletian equalled 5.35 grams × 12 = 64.2 grams of silver. One argenteus was theoretically based on the standard of the denarius of Nero, 3.4 grams × 0.935 fineness = 3.179 grams of pure silver. 64.2 divided by 3.179 gives 20 argentei to the solidus.

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A copper coin of one denarius was struck on a very limited scale; the sestertius did not exist any more.

It would have been a very good system had not the unit of reckoning been the denarius and had there not been a sizable inflation at the start of the new setup. And inflation was a phenomenon with which the Emperor did not know how to deal with.

The coin reform was combined with the spending of vast volumes of new coins. They were used as payment for the following projects:

- The army was greatly enlarged; it is suggested that 35 new legions were added to the 42 or 43 that existed at the beginning of Diocletian's reign (Jones 1964, 59).
- Many construction projects were started, but most of Diocletian's buildings were utilitarian and necessary: factories and mints, frontier forts, roads and bridges. The palace at Split and the bath at Rome were luxuries (Jones 1964, 66).
- Diocletian also built a solid administrative structure by giving each provincial governor a smaller area to control. This was important since a multitude of requisitions in kind had to be organized besides the old money taxes. This enlargement of the civil service induced Lactantius to state that the number of persons in receipt of state payment began to exceed the number of taxpayers.

In so far Diocletian and his co-regents followed the example of Augustus, Nero and Septimius Severus but unlike these predecessors they did not experience the favourable influences of their money reform and money issues. Instead a terrible inflation arose. We calculated an average annual inflation of 22.9% over the years 293–301. This means something like 10% or so in 294 mounting to 35% or more in 301. Inflation was an unknown phenomenon; it must have had very miserable consequences for the people and for the financial policy of the Emperors.

Because the denarius and sestertius were the basic measures of value, very soon after 293 the silver content of the argenteus must have had a higher value than the nominal value of this coin (20 denarii). The same must have been true of the gold value of the solidus (400 denarii in 293). Gresham's law no doubt gradually forced these coins out of circulation. For the solidus this meant that the coin did not again become a coherent part of the currency system as Diocletian intended, but remained a floating unit, its value measured by its gold content. This had long been the position of the aureus, probably since 190 A. D. The inflation firstly threatened the argenteus, which soon after 293 must have gradually been hoarded. Whether the circulation of the follis, with a much lower silver content also was compromised is not quite clear. We may conclude however that quite soon after its introduction the coherent currency system of Diocletian was seriously damaged. This must have been a great disappointment to the rulers of the Empire. To save at least part of the coin

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circulation they must have taken the emergency measure of raising the nominal value of the silver-containing coins: the argenteus and the follis. They took this measure late, as is evident from the size of the raise: 150%. This means that the nominal value of the argenteus became 50 denarii and that of the follis 12.5 denarii, which values are known from the inscription found in 1970 in Aphrodisias. Of this retariffing we do not have as yet any written evidence. Based on the scope of the inflation and the fact that in 301 a second retariffing was announced, we estimate that this first retariffing of Diocletian occurred in 297 or the beginning of 298.

The fact that the emergency measure of retariffing was taken late means that a great number of argentei must already have vanished from circulation. Circulating money was now predominantly copper. Problems with the silver coin may have induced the Emperors to enlarge the minting of solidi in the years 298/299 (Jahn 1976, 103). No doubt part of the effective money circulation was saved for a while by the retariffing, but at a price which Jungck (1976) points out: an increase of the nominal value of the coins means a tremendous enlargement at once of the effective quantity of money. The measure certainly will have boosted the inflation! A few years later the Emperors must have experienced that the retariffing of the silver coins had only brought temporary relief and new measures were again necessary to save the money circulation. Jungck calculates the urgency of new measures: just before September 301 the argenteus had a nominal value of 50 denarii, but its silver content had a value of 62.5 denarii (Jungck 1976, 27). We are well informed since 1970 about the second retariffing. An inscription found in that year tells that the nominal value of the argenteus was retariffed from 50 to 100 denarii on September 1, 301, while another coin – no doubt the follis – was retariffed from 12.5 to 25 denarii. The fact that the solidus is not mentioned indicates that this coin was no longer considered as a real interchangeable coin of a coherent currency system, but a floating unit of value. The inscription of Aphrodisias shows that the official value of the silver coins rose by 500% between 293 and 301; the general price level of 301 must therefore have been roughly five times that of 293. A conclusion which induces us to estimate the price index for 293 at one fifth of that of 301, that is 1400 (see Table paragraph 1).[18]

These developments must have been a great disappointment to Diocletian and his co-regents. The currency reform had failed. Corrective measures had

[18] The estimated price index of 1400 for 293 A. D. may be checked by the following approach: the argenteus of 293 is of a standard equal to that of the Neronian denarius of 64 A. D. For 64 an index of 75–85 was calculated. The argenteus equalled 20 denarii, which brings the index for 293 to 20 × 75 or 20 × 85 = 1500 – 1700. The deviation from 1400 seems acceptable given the uncertainties involved and the very general approach to the problem, the more so if we take into consideration that the price level between Augustus and Diocletian probably decreased somewhat if measured in the value of silver.

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been necessary already twice within eight years. The issue of additional money in order to enlarge the effective quantity of money did not result in enlarged prosperity for the people but just in the opposite: inflation and all ensuing miseries. Diocletian must finally have understood that the only way to stop the unfavourable developments was to end the inflation in the Empire. He tried to do so by forbidding to raise the prices above a certain level on penalty of death. This was laid down by Diocletian and his co-regents in the Edict of Maximum Prices which was issued in the last months of 301. For more than 1200 products and services a maximum price was fixed. The Preamble to the Edict gives the reasons for the issuing of maximum prices. It shows that the Emperors did not understand the origins of the inflation phenomenon. They believed that prices rose while the harvests were good because of the boundlessness of human greed!

11. The causes of the hyperinflation under Diocletian

With the formula of reforming the currency system to a sound, coherent and flexible entity followed by a – "Keynesian" – spending of much money that mobilized available labour and land, the Emperors Augustus, Nero and Septimius Severus achieved stability of the economy and an increase in prosperity. Probably also Caracalla and Aurelian tried to apply the same formula but their periods of reign were too short to show a result. The money reform of Caracalla at least failed.

But for Diocletian, who also tried to apply the formula, the developments are clear. A currency reform was introduced in 293 or a little later, after which the Emperor still ruled twelve years. Long enough to experience that this time the formula did not work. Instead of an increase in prosperity it brought a hyperinflation with all negative consequences involved. The question should be put forward why the economic developments now were different from those of the past.

It firstly is important to conclude that the inflation was not the result of a shortage of consumption goods. The Preamble to the Edict of Prices is clear about that. In the second place some authors analysed that as far as prices were expressed in gold, there was no inflation at all (Jones 1974, 207). The causes of the hyperinflation must therefore be found in the monetary policy of the Emperors.

An analysis of the Roman monetary policy reveals that the following seven points are of importance for the explanation of the hyperinflation under the reign of Diocletian and his co-regents.

Firstly, when Diocletian became Emperor in 284, the state of inflation had existed already for some decades. For 284 an inflation rate of 5% may be estimated. Diocletian, being unfamiliar with the inflation phenomenon, did

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not take measures with, or before his currency reform of 293 to stop the inflation spiral. Of course he did not have many options for such measures, but a move to a gold standard currency system, with the introduction of a gold coin as unit of reckoning, could have been a solution.

Secondly, the denarius and the sestertius, the quarter denarius, were since 217 B. C. the units of reckoning in the Roman world. Notwithstanding this fact the financial policy – certainly from Nero to Pertinax (64–193 A. D.) – aimed at maintaining the standard of the aureus. The standard of the denarius, the reckoning unit, was manipulated to maintain the value of the aureus and to conserve the exchange rate between the two coins. For this manipulation with the standard of the unit of reckoning a price had to be paid in the end.

Thirdly, the relation between the gold value of the aureus and the silver value of 25 denarii was deliberately abandoned by Septimius Severus after 193. He must have tried to make the denarius into a token, subsidiary coin. From that time on the denarius floated detached from the aureus and in fact also independent of the silver value, because the fineness considerably fluctuated since 193 and therefore gradually lost its monetary significance. Caracalla in 215 tried presumably to restore the relation between aureus and denarius, but failed. Diocletian in his currency reform moreover incorporated a restored relation between a silver coin, the argenteus, and the gold coin, that probably by then was called solidus. However, it was too late to restore the bimetallic currency system.

Fourthly, Diocletian was forced to increase the nominal value of the argenteus and the most important coin, the follis, in 297 or 298 by 150% and in 301 by 100%. This meant that the effective quantity of money increased significantly at these moments of retariffing. Since nothing changed in the supply of goods and services then, prices must have rocketed.

Fifthly, the strong inflation forced many silver and gold coins into hoards. A shortage of those coins was soon felt. This induced the government to mint more copper and bronze coins and to debase the follis. The currency system of Diocletian was therefore soon seriously damaged. Exchange of copper coins for the argenteus was hardly possible. The coin system became effectively a bronze based system as had been the case in the early third century B. C.

Sixthly, the government of Diocletian annually minted and distributed large quantities of copper coins and withdrew none from circulation by taxation (Jones 1964, 442).

And last, but certainly not least, is it important to ascertain the fact that at the time of Augustus, Nero and Septimius Severus there were production factors available, land and labour, to be mobilized. Augustus' reign was at the end of a civil war that lasted more than half a century. The Empire came to a rest, soldiers were demobilized, land was available. Nero's rule was at the end of a deflationary period that lasted over 70 years, while Septimius Severus

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reigned after a long period in which the country was struck by the plague, the population increased again and abandoned lands could be re-used. Diocletian also ruled after half a century of civil wars, but the situation was different. Diocletian created order and justice by perfectly organizing an economy in kind. Annual budgets with requisitions and payments in kind operated the system, which brought hereditary castes of workers. This prevented available labour to move in the direction that money was spent. Added to that there are reasons to believe that a shortage in manpower had developed within the Roman Empire. The disorders of the troubled period 235 to 284 were especially bound to have an extremely unfavourable effect upon the population, both rural and urban (Boak 1955, 112). There was maybe, not a shortage of day to day consumption goods facing the rising quantities of money, but there were no production factors available to move to a more productive application. The economy had become rigid, there was no possibility left to increase individual prosperity. A multiplier effect of spending money was impossible, society remained as it was. The result was that more money was placed opposite the same number of goods and services. Even an improvement in the distribution of labour within the country could not be realized.

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